

KHASAB PORT, Oman

WMO: 412400

Lat: 26.211N

Lon: 56.244E

Elev: 13

StdP: 14.69

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.5	62.6	25.9	19.9	89.7	30.2	24.4	86.6	20.9	70.5	18.6	70.3	6.0	140	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	109.4	70.5	106.8	71.6	104.5	72.6	86.7	93.2	85.8	93.4	84.8	93.4	10.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
85.1	184.7	92.2	83.8	177.1	92.2	82.7	170.4	92.0	51.8	93.2	50.6	93.0	49.5	93.5	90.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.7	15.0	57.6	114.0	1.8	1.8	56.3	115.3	55.2	116.3	54.2	117.3	52.9	118.6		
(5)			45.1	88.6	1.9	1.3	43.7	89.6	42.6	90.3	41.5	91.1	40.1	92.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	84.5	69.6	71.6	76.4	84.6	92.0	94.5	96.6	96.0	93.1	87.9	79.1	71.8	(6)
(7)		DBStd	10.63	3.37	3.98	4.73	5.60	4.81	4.39	3.29	3.17	3.04	3.23	3.70	3.46	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	9	5	2	0	0	0	0	0	0	0	0	0	2	(9)
(10)		CDD50	12593	608	606	817	1038	1302	1336	1444	1427	1293	1175	872	675	(10)
(11)		CDD65	7124	147	187	352	588	837	886	979	962	843	710	422	211	(11)
(12)		CDH74	100422	323	683	2631	7541	13108	14412	16479	16215	13500	10444	4212	874	(12)
(13)		CDH80	63610	12	77	731	3804	8647	10092	12015	11751	9180	6028	1212	61	(13)
(14)	Wind	WSAvg	6.5	6.4	6.7	6.4	6.3	6.4	6.5	7.5	7.9	6.7	5.1	5.6	6.2	(14)
(15)	Precipitation	PrecAvg	6.5	1.5	1.3	1.3	0.2	0.0	0.0	0.0	0.0	0.1	0.4	1.3		(15)
(16)		PrecMax	14.6	7.4	5.0	5.2	1.8	0.1	0.1	0.7	0.1	0.1	1.2	3.0	8.9	(16)
(17)		PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.7	1.8	1.3	1.6	0.4	0.0	0.0	0.1	0.0	0.0	0.3	0.7	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.7	85.4	95.6	102.3	111.7	112.7	111.8	110.7	107.1	102.1	91.8	84.6	(19)
(20)			MCWB	61.9	63.8	62.7	66.3	68.7	68.1	71.9	71.0	69.6	67.8	68.7	63.5	(20)
(21)		2%	DB	79.0	82.5	90.5	99.8	106.5	109.1	108.0	107.8	104.2	98.8	89.1	81.8	(21)
(22)			MCWB	62.2	64.0	63.2	66.3	67.7	68.9	74.3	72.6	71.5	68.8	69.2	63.4	(22)
(23)		5%	DB	77.3	80.0	86.3	96.9	103.7	105.7	105.0	105.1	101.1	96.0	87.2	79.9	(23)
(24)			MCWB	63.2	63.9	64.0	65.6	67.4	69.8	76.2	75.2	73.0	70.7	70.4	64.1	(24)
(25)		10%	DB	75.7	77.9	83.5	94.1	100.9	102.4	102.4	102.3	98.7	93.9	85.6	78.4	(25)
(26)			MCWB	63.4	64.2	64.5	65.9	68.1	71.2	78.4	77.0	75.7	72.7	70.5	64.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	71.5	73.8	79.1	84.3	87.0	88.2	87.2	87.1	83.8	77.2	72.0	(27)
(28)			MCD B	75.9	78.1	80.4	85.9	88.9	93.0	92.5	94.3	92.2	90.5	84.8	78.6	(28)
(29)		2%	WB	68.1	69.9	72.4	76.8	82.4	85.6	87.0	86.1	85.1	82.2	75.8	70.0	(29)
(30)			MCD B	75.1	77.1	79.3	84.3	88.1	92.4	93.1	94.4	92.0	90.5	84.0	77.1	(30)
(31)		5%	WB	66.7	68.8	71.0	75.3	80.9	84.5	86.0	85.2	83.7	80.8	74.5	68.3	(31)
(32)			MCD B	74.4	76.1	78.5	83.6	88.4	91.9	93.4	94.2	92.1	89.6	83.4	76.0	(32)
(33)		10%	WB	65.5	67.5	69.9	73.9	79.5	83.4	85.0	84.2	82.8	79.6	73.3	66.9	(33)
(34)			MCD B	73.5	75.1	77.9	83.6	88.6	91.6	94.0	94.1	92.0	88.9	83.1	75.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.0	9.5	11.0	11.7	10.4	9.2	9.1	9.2	10.0	9.7	10.3	(35)
(36)			MCDBR	11.5	11.2	13.9	14.9	17.5	16.8	14.3	14.0	13.8	13.5	11.5	12.7	(36)
(37)			MCWBR	7.1	7.4	8.2	8.3	9.4	11.5	9.8	9.2	10.0	10.4	8.4	8.8	(37)
(38)		5% WB	MCDBR	9.5	8.1	9.4	11.0	10.5	9.0	7.8	7.8	7.8	8.4	9.3	9.7	(38)
(39)			MCWBR	7.2	6.8	7.7	9.2	8.3	8.4	7.7	7.7	8.0	8.6	8.2	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.435	0.497	0.553	0.579	0.581	0.637	0.720	0.676	0.594	0.514	0.449	0.429	(40)	
(41)		taud	2.178	1.996	1.858	1.822	1.822	1.748	1.639	1.721	1.842	1.992	2.168	2.197	(41)	
(42)		Ebn at Noon	254	246	239	237	236	222	204	212	226	238	247	250	(42)	
(43)		Edh at Noon	41	53	64	68	68	73	81	75	64	53	42	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1255	1495	1819	2128	2396	2422	2239	2149	1975	1720	1369	1194	(44)	
(45)		RadStd	82	120	127	119	56	55	57	57	34	31	42	77	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.83	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	+250	+241

Nomenclature: See separate page